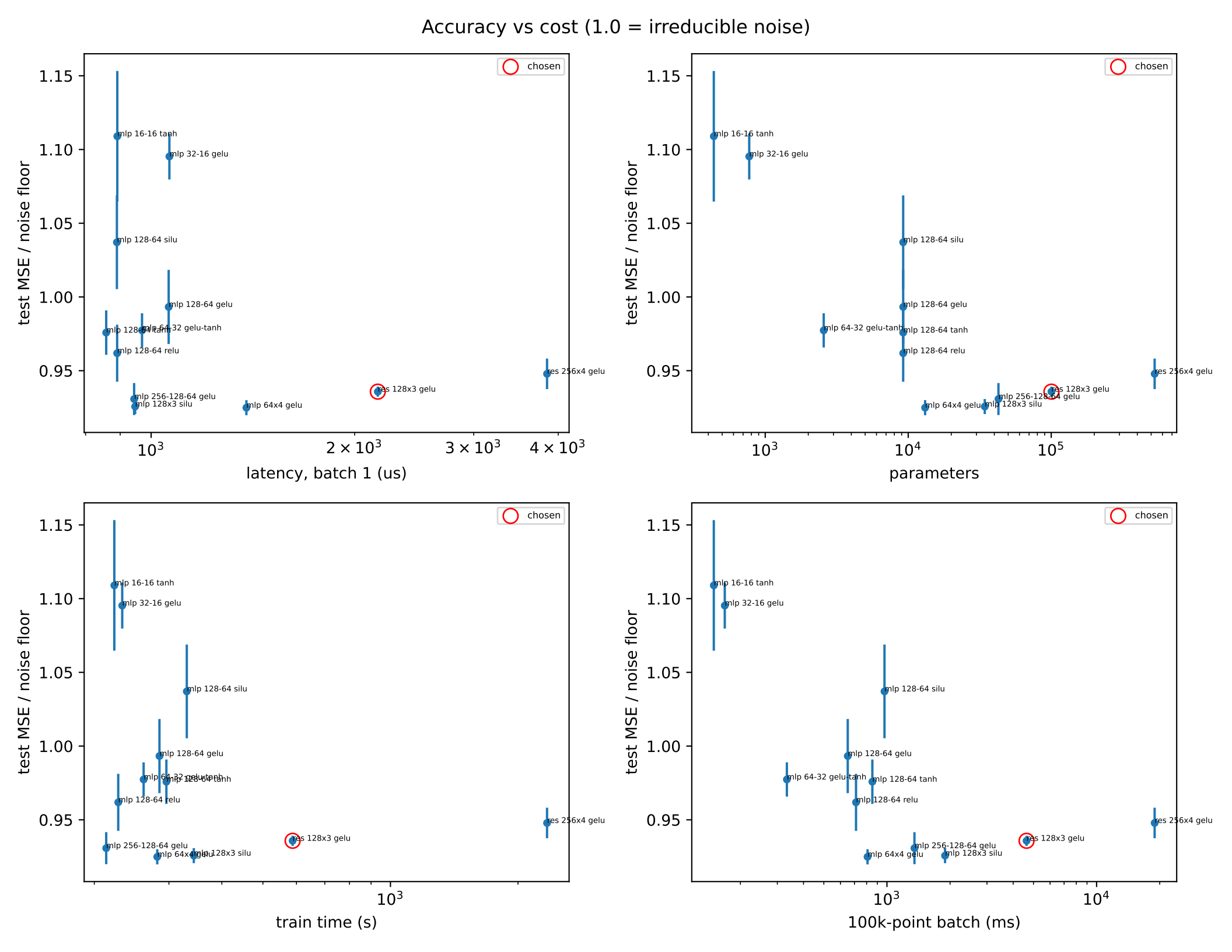
4D DNN SURROGATE - ARCHITECTURE SEARCH REPORT

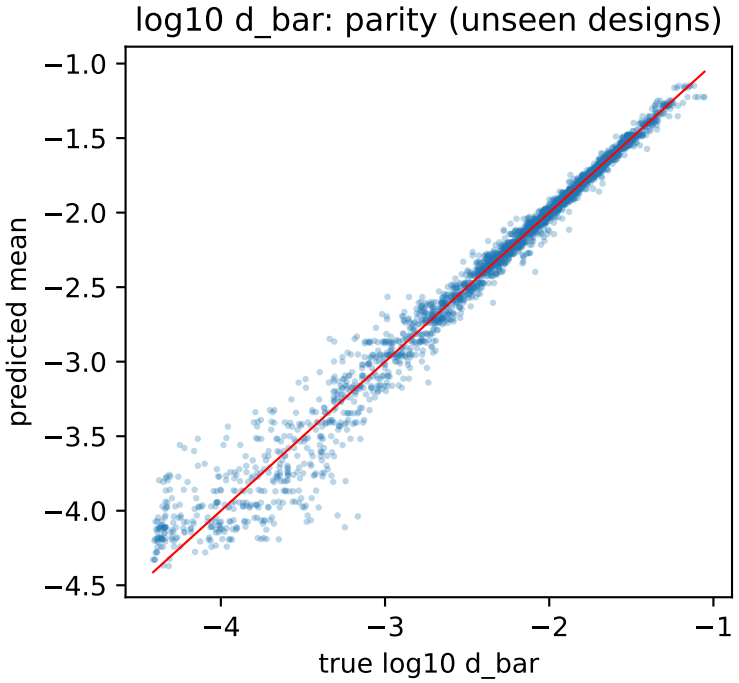
GOAL: replace the GP surrogate in ABC-MCMC. For theta=(log10 p1, log10 p2, tau) predict the
joint distribution of the two ABC features of the J=100 cultures: d_bar (mean of (X/Z)^(1/4)) and s (its sd),
fast and with an honest spread.   Data: 50,000 rows (10,000 LHS designs x 5 reps), paper box, tp=20, delta free.
30.1% of rows have d_bar = 0, so the model is a hurdle model:
P(d_bar=0) (logistic head) + Gaussians on (log10 d_bar, log10 s) for nonzero (mean, log-variance heads).
Split BY DESIGN point 80/10/10 (unseen theta), 60 fits = 12 architectures x 5 seeds, 16 cores, 44.1 min.
Noise floor (irreducible within-design variance), test: log10 d_bar 1.13e-02, log10 s 1.23e-02.

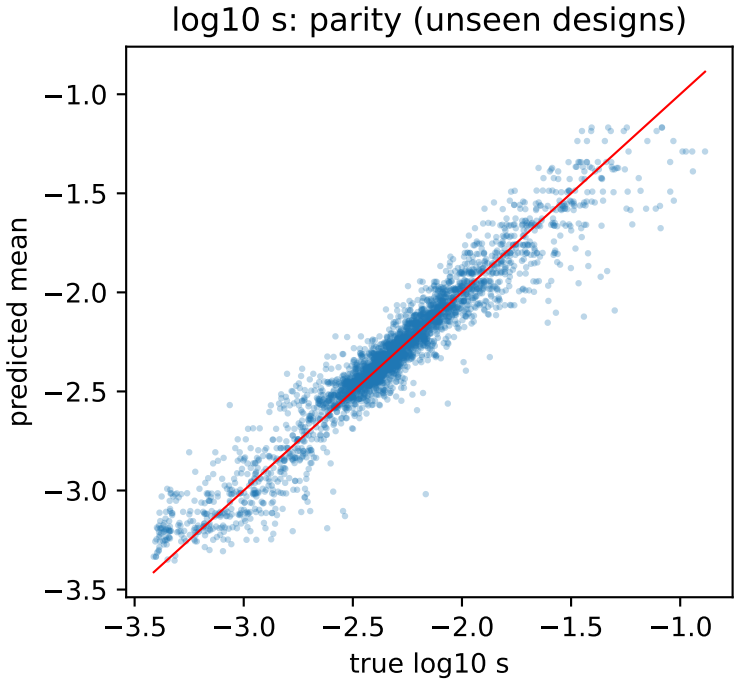
SELECTION (rule fixed in advance): best mean validation loss = res_128x3_gelu;
chosen = cheapest model within 1 SE (0.0017) of it = res_128x3_gelu
Chosen model (seed 0, saved to surrogate_3d.pt) on held-out TEST designs:
  nonzero part (d_bar | s): MSE = 0.93 | 0.94 x noise floor, RMSE 0.1026 | 0.1075 (log10),
                95% coverage 0.943 | 0.953 after conformal (raw 0.945 | 0.947)
  zero part   : log-loss 0.0540, Brier 0.0172, AUC 0.9979
GP baseline (GPS-ABC style, 300 pts, nonzero part only, cannot model the zero atom):
  log10 d_bar only: RMSE 0.1634 = 2.37 x floor, 95% coverage 0.914, fit 16s, query 2881 us

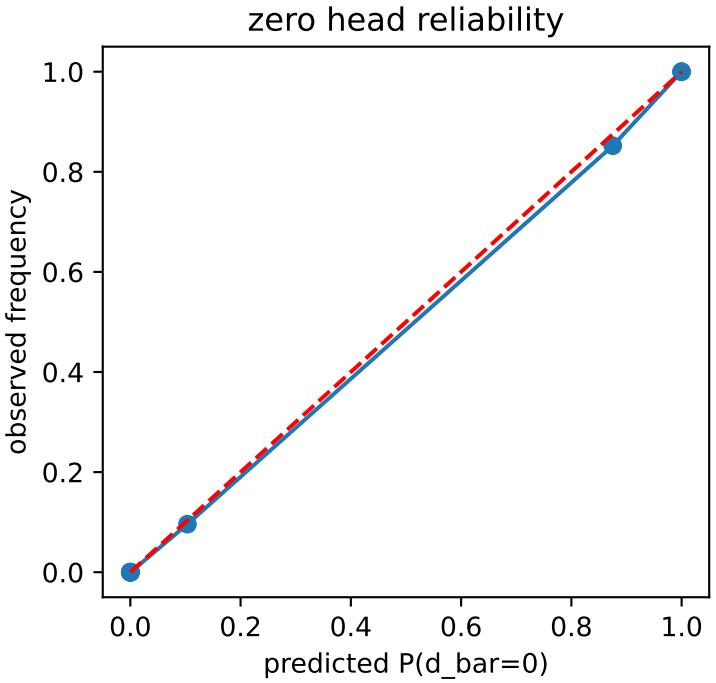
Mean over 5 seeds (sorted by validation loss; test_* are held-out designs; latency = batch 1, microseconds)

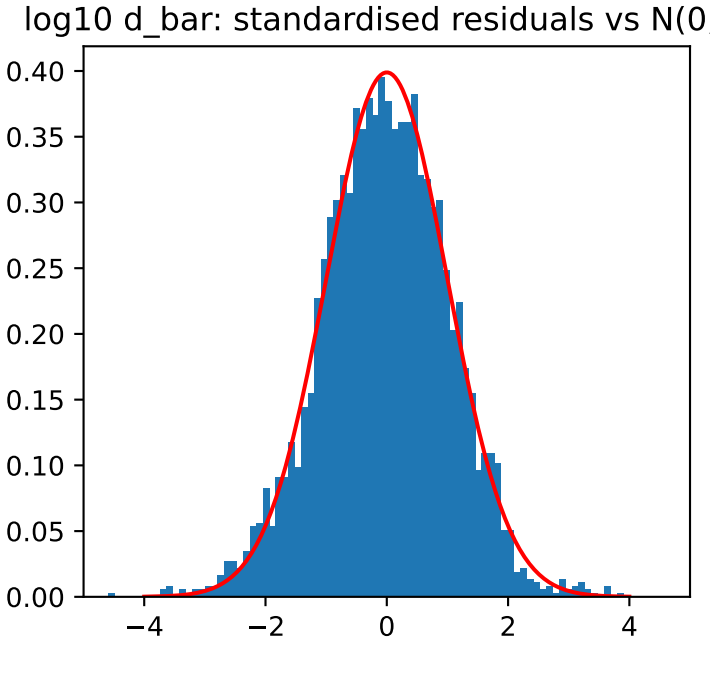
arch	params	val_loss	test_x_floor	test_rmse	test_nll	test_ll0	test_auc0	test_cov95_conf	latency_us	batch1e5_ms	train_s	x_floor_sd
res_128x3_gelu	100,357	-3.342	0.9358	0.105	-2.901	0.05334	0.9979	0.9483	2164	4635	587.6	0.003434
mlp_64x4_gelu	13,125	-3.337	0.9249	0.1044	-2.907	0.05337	0.9979	0.9476	1383	808.1	281.6	0.005114
mlp_128x3_silu	34,309	-3.328	0.9257	0.1045	-2.896	0.05393	0.9979	0.9468	947	1893	343.5	0.005102
mlp_256-128-64_ge	42,757	-3.321	0.9308	0.1047	-2.887	0.0536	0.9979	0.9474	943.8	1352	213.4	0.01085
res_256x4_gelu	528,901	-3.296	0.9479	0.1057	-2.856	0.05415	0.9978	0.9486	3851	1.892e+04	2342	0.01035
mlp_128-64_tanh	9,221	-3.292	0.9758	0.1072	-2.846	0.05526	0.9978	0.9467	858.2	851.8	295.9	0.01506
mlp_128-64_gelu	9,221	-3.289	0.9932	0.1081	-2.853	0.05449	0.9978	0.9455	1062	650.2	285	0.02514
mlp_64-32_gelu-ta	2,565	-3.284	0.9773	0.1073	-2.836	0.05535	0.9978	0.9436	969.5	333.2	261.5	0.0116
mlp_128-64_silu	9,221	-3.275	1.037	0.1105	-2.836	0.05472	0.9978	0.9448	890	973.3	330.6	0.03177
mlp_128-64_relu	9,221	-3.27	0.9618	0.1065	-2.831	0.05442	0.9978	0.9474	890.7	712	227.8	0.01934
mlp_32-16_gelu	773	-3.193	1.095	0.1135	-2.755	0.05688	0.9977	0.947	1064	168.4	232.8	0.01558
mlp_16-16_tanh	437	-3.15	1.109	0.1143	-2.692	0.05959	0.9975	0.9447	891.2	149.3	223	0.04429

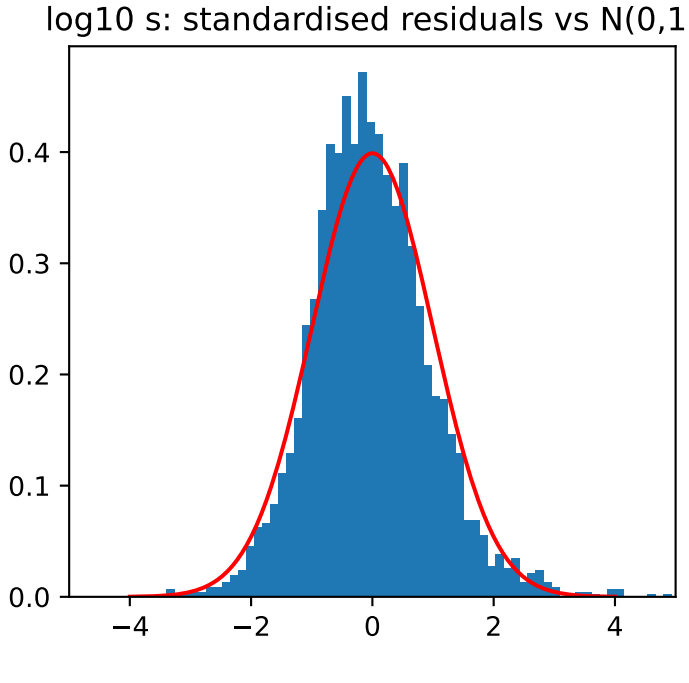

Accuracy vs cost (1.0 = irreducible noise)
test MSE / noise floor
latency, batch 1 (us)
parameters
train time (s)
100k-point batch (ms)
chosen
mlp 16-16 tanh
mlp 32-16 gelu
mlp 128-64 silu
mlp 128-64 gelu
mlp 64-32 gelu-tanh
mlp 128-64 tanh
mlp 128-64 relu
mlp 256-128-64 gelu
mlp 128x3 silu
mlp 64x4 gelu
res 128x3 gelu
res 256x4 gelu

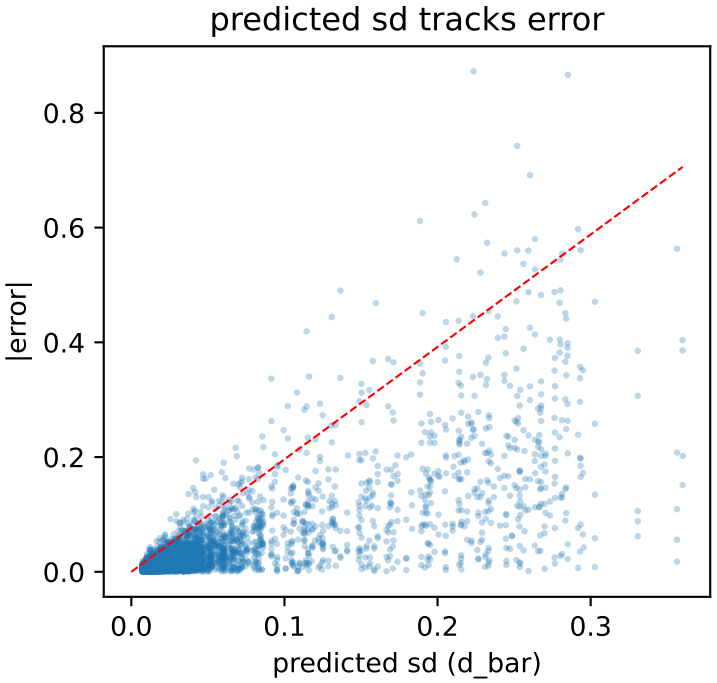

log10 d_bar: parity (unseen designs)
predicted mean
true log10 d_bar


log10 s: parity (unseen designs)
predicted mean
true log10 s

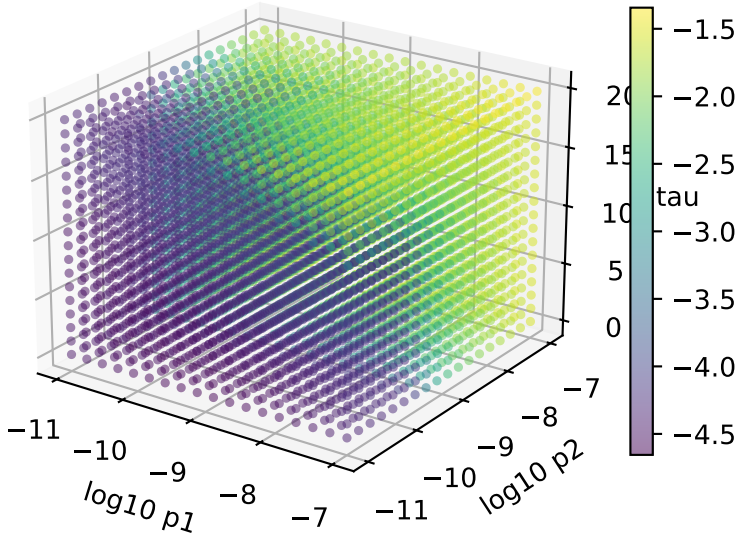

zero head reliability
observed frequency
predicted P(d_bar=0)


log10 d_bar: standardised residuals vs N(0,1)


log10 s: standardised residuals vs N(0,1)


predicted sd tracks error
|error|
predicted sd (d_bar)

predicted mean log10 d_bar (nonzero), delta=1 slice



predicted P(d_bar = 0)

